PhotonEdge

Beijing Hengdingguang Technology Co., Ltd.

Precision Optical Components

PRODUCT CATALOG

Precision in Optics, Excellence in Innovation

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About Us

Beijing Hengdingguang Technology Co., Ltd. is a professional company integrating R&D, production, and sales of optical components and products. As the core operator of the PhotonEdge brand, we are committed to providing global customers with high-quality precision optical components and comprehensive solutions.

Core Advantages

• **Complete Product Line:** Full range of optical elements including lenses, mirrors, windows, prisms, filters, waveplates, beamsplitters, and polarizers

• **Premium Material Selection:** K9 (BK7), Fused Silica, Zinc Selenide, Germanium, Silicon, Calcium Fluoride, Sapphire, and other optical materials

• **Advanced Manufacturing Equipment:** Precision optical component production lines and advanced processing and testing equipment

• **Professional Technical Team:** High-quality R&D technicians, senior management personnel, and experienced production and sales staff

• **Flexible Customization Services:** Large ready inventory, OEM custom processing available to meet diverse customer needs

Application Fields

• Medical Instruments

• Optoelectronic Instruments

• Educational Instruments

• Fiber Optic Communication

• Military Equipment

• Construction Surveying

• Laser Processing

Quality Policy

Customer First, Quality Priority, Technology Innovation, Perfect Service

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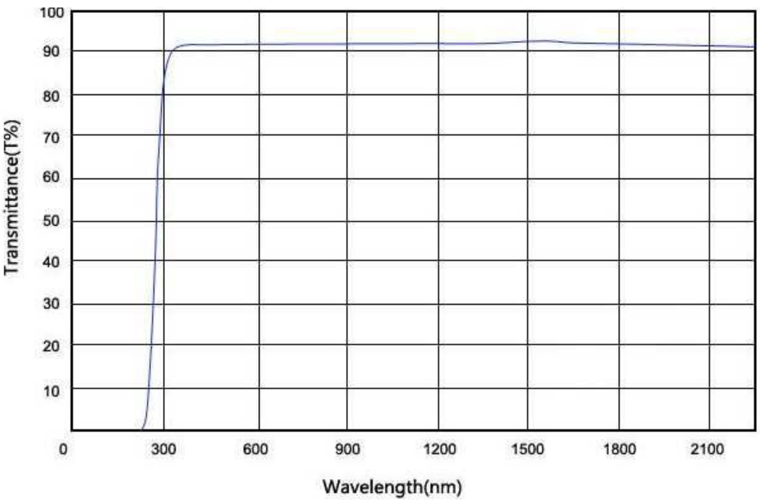
I. K9 (BK7)

K9 is a commonly used optical glass material with a wide range of applications. It can be used for processing optical components such as lenses, prisms, windows, and mirror substrates.

K9 (BK7) Physical Properties

Name	Refractive Index Nd	Transformation Temp. Tg [°C]	Thermal Expansion Coeff. [$\times 10^{-7}$/°C]	Microhardness HK [$\times 10^{7}$ Pa]	Density [g/cm³]	Bubble Grade
K9	1.51637	569	75 (20–120°C) 81 (20–300°C)	570	2.53	A

Transmission Curve



II. Fused Silica Materials (JGS1/JGS2/JGS3)

Optical quartz glass windows can withstand high temperatures and pressure. Mainly applied in: special light sources, optical instruments, optoelectronics, military industry, metallurgy, semiconductors, optical communication, and other fields. Maximum service temperature: 1200°C, softening temperature: 1730°C.

JGS1 (Far UV Optical Quartz Glass): High-purity synthetic fused silica with excellent UV transmission, especially in the short-wave UV region. Its transmission far exceeds all other glasses in this region. The transmittance at 185 nm can reach 90%, making it an excellent optical material for the 185–2500 nm wavelength range.

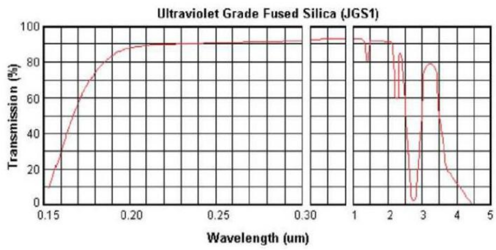
JGS2 (UV Optical Quartz Glass): Oxyhydrogen-fused optical quartz glass. It is an excellent material for the 220–2500 nm wavelength range.

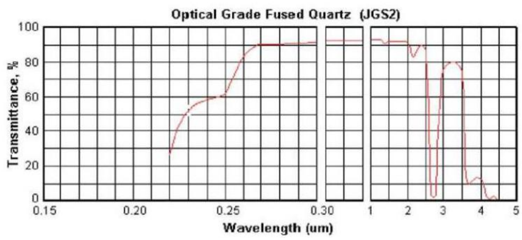
JGS3 (IR Optical Quartz Glass): Features high IR transmission with transmittance above 85%. Suitable for the 260–3500 nm wavelength range.

Fused Silica Physical Properties

Name	Refractive Index Nd	Transformation Temp. Tg [°C]	Short-term Service Temp. [°C]	Long-term Service Temp. [°C]	Density [g/cm³]	Hot Working Temp. [°C]
Fused Silica	1.4585	1280	1300	1100	2.2	1750–2050

Transmission Curve





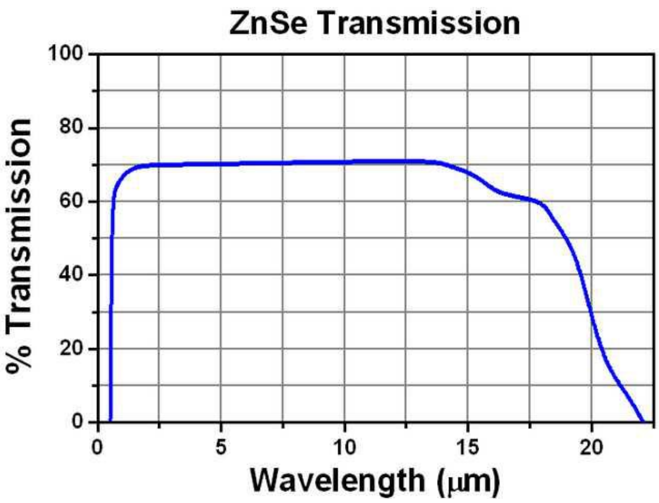
III. Zinc Selenide (ZnSe)

Zinc Selenide (ZnSe) is a chemically inert material characterized by high purity, excellent environmental adaptability, and easy processability. It features low light transmission loss and excellent transparency. It is the preferred material for high-power CO_2 laser optical components. Due to its uniform refractive index and excellent consistency, it is also an ideal material for protective windows and optical elements in Forward-Looking Infrared (FLIR) thermal imaging systems. Additionally, this material is widely used in medical and industrial thermal radiation measurement instruments and IR spectrometers for windows and lenses.

Zinc Selenide Physical Properties

Name	Refractive Index ηd	Melting Point [°C]	Density [g/cm³]	Bending Strength
ZnSe	2.48	1525	5.327	55 MPa

Transmission Curve



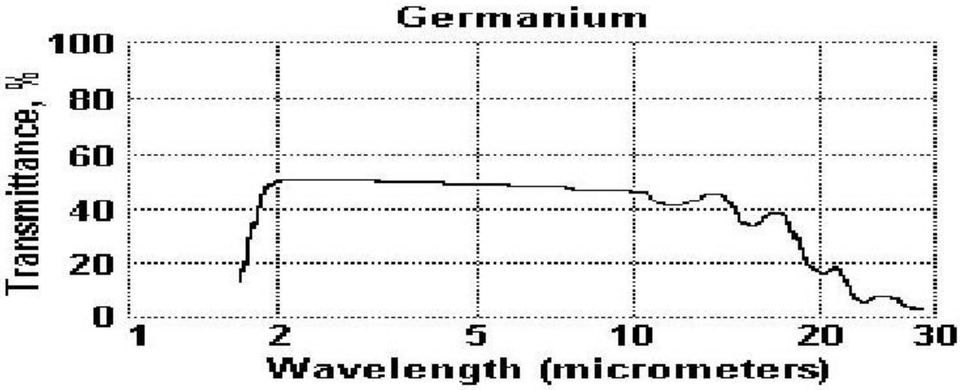
IV. Germanium (Ge)

Germanium (Ge) is a chemically inert material and one of the most commonly used infrared optical materials. It features high hardness, excellent thermal conductivity, and is insoluble in water. It is widely used in infrared imaging systems and infrared spectrometer systems.

Germanium Physical Properties

Name	Refractive Index ηd	Melting Point [°C]	Density [g/cm³]	Thermal Expansion Coeff.
Ge	4.02	936	5.33	6.1×10^{-6} @ 298 K

Transmission Curve



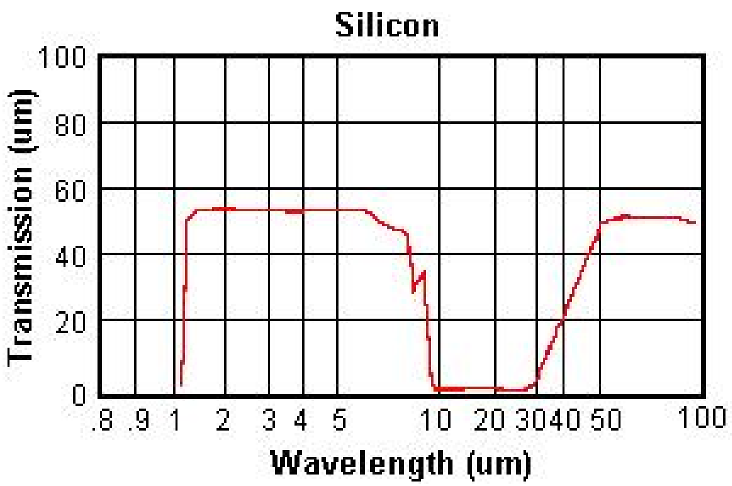
V. Silicon (Si)

Silicon (Si) is a chemically inert material with high hardness and insoluble in water. It exhibits excellent transmission in the 1–7 µm wavelength range. Additionally, it demonstrates excellent transmission in the far-infrared range of 50–300 µm, a characteristic not possessed by other infrared materials.

Silicon Physical Properties

Name	Refractive Index ηd	Melting Point [°C]	Density [g/cm³]	Thermal Expansion Coeff.
Si	3.47	1420	2.33	4.15×10^{-6} @ 298 K

Transmission Curve



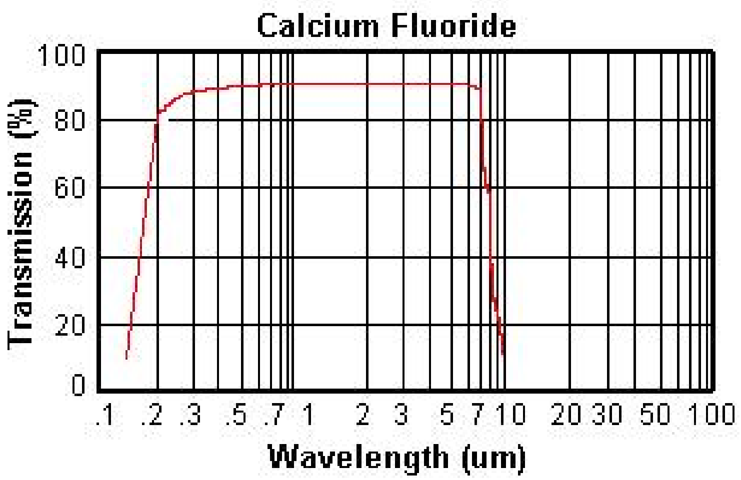
VI. Calcium Fluoride (CaF_2)

Calcium Fluoride (CaF_2) exhibits excellent transmission across UV, visible, and IR wavelength ranges. It is widely used in lasers, IR optics, UV optics, and high-energy detector technologies.

Calcium Fluoride Physical Properties

Name	Refractive Index ηd	Melting Point [°C]	Density [g/cm³]	Thermal Expansion Coeff.
CaF_2	1.44	1360	3.18	18.85×10^{-6} @ 298 K

Transmission Curve



VII. Sapphire (Al_2O_3)

Sapphire (Al_2O_3) is composed of aluminum oxide with extremely high hardness, classified as a superhard material. Sapphire has a very wide optical transmission band, exhibiting excellent transmission from near-UV (190 nm) to mid-infrared wavelengths. Therefore, it is extensively used in optical components, infrared devices, high-intensity laser lens materials, and mask materials. It features high sound velocity, high temperature resistance, corrosion resistance, high hardness, high transparency, and a high melting point (2045°C). As a challenging material to process, it is commonly used as material for optoelectronic components.

Sapphire Physical Properties

Name	Refractive Index ηd	Melting Point [°C]	Density [g/cm³]	Thermal Expansion Coeff.
Sapphire	1.44	2045	3.98	8.4×10^{-6}/K

Transmission Curve